GENERAL CERTIFICATE OF EDUCATION BOARD

General Certificate of Education Examination

JUNE 2024 **ADVANCED LEVEL**

Subject Title	**Pure Mathematics With Mechanics**
Subject Code No.	**0765**
Paper No.	**Paper 2**

Duration: Three Hours.

Full marks may be obtained for answers to ALL questions.

Mathematical formulae Booklets published by the Board are allowed.

In calculations, you are advised to show all the steps in your working, giving the answer at each stage.

Calculators are allowed.

Start each question on a fresh page.

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1. (i) Express $\frac{4x + 3}{(2x - 1)(3x + 1)}$, $x \in \mathbb{R}$, $x \neq -\frac{1}{3}, \frac{1}{2}$ in partial fractions. **(3 marks)**

(ii) If the roots of the quadratic equation $x^2 + (k + 1)x + k = 0$ are α and β, find the values of the real constant k for which $\alpha = 2\beta$. **(5 marks)**

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2. The polynomial $f(x)$, where $f(x) = ax^3 - x^2 + bx + 6$, has a factor $(x + 2)$. $f(x)$ leaves a remainder 10 when divided by $(x + 1)$.
Find
(a) the values of the constants a and b,
(b) the values of x for which $f(x) = 0$. **(8 marks)**

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3. The table below shows approximate values of a variable y corresponding to certain values of another variable x.

x	10	20	35	50	60	80
y	3.47	5.50	10.47	20.89	33.11	79.43

By drawing a suitable linear graph, verify that these values of x and y satisfy approximately a relationship of the form $y = ab^x$.

Using your graph, find approximate values of the constants a and b, giving the answers correct to 2 decimal places. **(10 marks)**

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4. (i) The first and last terms of an arithmetic progression are 7 and 43 respectively.
The sum of the terms of the progression is 250.
Find the number of terms and the common difference of the progression. **(3 marks)**

(ii) Obtain the first four terms of the binomial expansion of $\left(1 - \frac{3}{2}x\right)^{-2}$ and find the set of values of x for which the expansion is valid.
Hence, deduce the first four terms of the binomial expansion of $(2 - 3x)^{-2}$. **(7 marks)**

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5. (i) The parametric equations of a curve are given by

$$x = 1 + t, \quad y = 2 + \frac{1}{t}, \quad t \neq 0 \text{ is a parameter.}$$

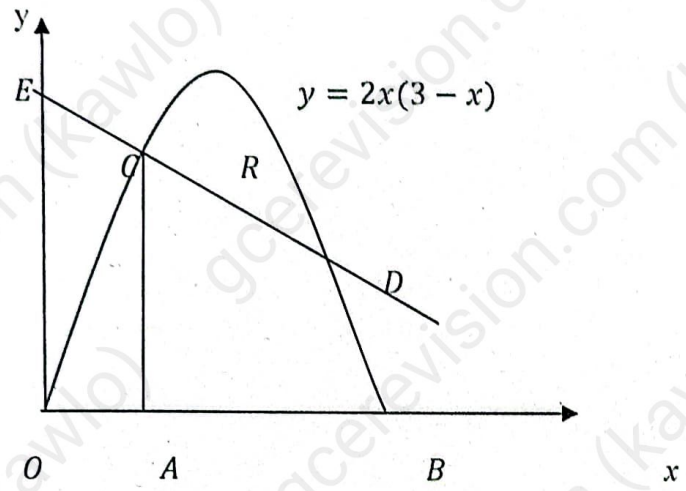
Show that an equation of the normal to the curve at the point where $t = 2$ is

$$8x - 2y - 19 = 0.$$

Find the value of t at the point where this normal meets the curve again. **(6 marks)**

(ii) A new primary school intends to recruit a head teacher and 7 other teachers.
There are 3 male and 2 female applicants for the position of head teacher, 4 male and 6 female other applications for the teaching positions.
Find the number of different ways of selecting the staff of 8 if gender equity must be respected. **(7 marks)**

6.



The curve with the equation $y = 2x(3 - x)$ crosses the x-axis at O and B.

(a) State the coordinates of the point B. **(1 mark)**

A straight line,which crosses the y-axis at the point E with coordinates (0, 5), meets the curve at the points C and D, as shown in the diagram above. The coordinates of the point D are (k, k).

(b) Show that the value of k is $\frac{5}{2}$. **(2 marks)**

(c) Find an equation of the straight line passing through D and E. **(3 marks)**

(d) Show that the x-coordinates of the point C is 1. **(3 marks)**

(e) Calculate the area of the region R bounded by the curve $y = 2x(3 - x)$ and line DE. **(4 marks)**

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7. (i) A complex number z satisfies the equation $\frac{z}{z+2} = 2 - i$.

Express z in the form $a + bi$, where a and b are real constants. **(4 marks)**

(ii) The vector equations of two lines are given by

$\mathbf{r} = 3\mathbf{i} + 6\mathbf{j} + \mathbf{k} + \lambda(2\mathbf{i} + 3\mathbf{j} - \mathbf{k})$ and $\mathbf{r} = 3\mathbf{i} - \mathbf{j} + 4\mathbf{k} + \mu(\mathbf{i} - 2\mathbf{j} + \mathbf{k})$.

(a) Find the vector normal to the plane containing both lines

(b) Obtain the Cartesian equation of the plane containing both lines **(3, 3, marks)**

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8. (i) The equation of a line is $3y = x + 5$ and the equation of a circle is $x^2 + y^2 - 6x = 2y - 15 = 0$.

Show that the line is a chord to the circle. **(5 marks)**

(ii) Show that the equation $x^2 - 3x + 1 = 0$ has a root between $x = 2$ and $x = 3$.

Taking $x = 3$ as a first approximation to this root, use the Newton – Raphson method to obtain a second approximation to the root. **(5 marks)**

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9. (i) Solve the differential eqution $x\frac{dy}{dx} = y(y+1)$, given that $y = 4$, when $x = 2$ and express the solution in the form $y = f(x)$. **(7 marks)**

(ii) Using a suitable substitution, show that

$$\int_0^2 x\sqrt{1+x^2}\,dx = \frac{1}{3}(5\sqrt{5}-1).$$

(5 marks)

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10. (i) Find the values of R and α , $R > 0$ and $0° < \alpha < 90°$, for which
$4\cos\theta + 3\sin\theta \equiv R\cos(\theta - \alpha)$
Hence, obtain the greatest value of the expression $\frac{1}{4\cos\theta + 3\sin\theta + 6}$, and give the corresponding value of θ between $0°$ and $360°$. **(6 marks)**

(ii) Using De Moivre's theorem, write down the equivalent form of $\cos 3\theta + i\sin 3\theta$.
Hence show that
$\sin 3\theta \equiv 3\sin\theta - 4\sin^3\theta$. **(6 marks)**

STOP

GO BACK AND CHECK YOUR WORK